

Khwaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Spring Semester 2019

Program: B.Sc. in CSE (Batch: 8th) 1st Year 1st Semester

Course Title: Physics – I

Course Code: PHY 1101

Total Time: 2 hours

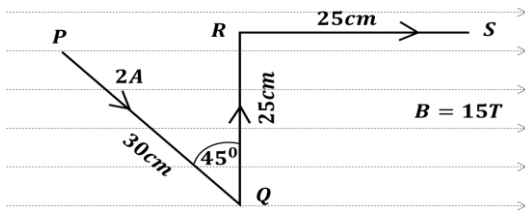
Full Marks: 40

Part-B

[Answer the question No. 1 and any four (04) from the rest] **5 × 7 = 35**

1. See the AC signal carefully and answer the questions that follows –
Two atoms ($m_1 = 2.6 \times 10^{-26} \text{ kg}$ and $m_2 = 5.7 \times 10^{-26} \text{ kg}$) are bound together with a force and they are oscillating about their equilibrium position. This oscillation can be replaced by a single particle SHM.
 - a) Define SHM. 1
 - b) Find the reduced mass of the above system. 3
 - c) Justify the last statement of the stem. 3
2.
 - a) Illustrate the damped oscillation graphically. 1
 - b) Draw the displacement vs total energy curve for SHM. 2
 - c) The displacement of a progressive wave at any instant of time is – $y = 0.1 \sin 2\pi(340t - 0.15)$, Calculate (i) the amplitude (ii) wave length (iii) frequency and (iv) time period. 4
3.
 - a) Define Lorenz force. 1
 - b) Write down the conclusion of the Oersted's Experiment. 2
 - c) Find the force of each segment of the wire: 4

Conducting wire in a magnetic field


4.
 - a) Define 1Henry. 1
 - b) Explain the application of Faraday's law in dynamo. 2
 - c) A flux of 0.5mWb is produced by a coil of 900 turns wound on a ring with a current of 3A in it. Calculate i) the self-inductance of the coil. ii) the emf induced in the coil when a current of 5A is switched off, assuming the current to fall to zero in 1 millisecond and iii) the mutual inductance of a second coil of 600 turns is uniformly wound over the first coil. 4
5.
 - a) State and explain the Ampere's law. What is the critical condition of it? 3
 - b) A solenoid 0.5m long has 2000 turns. The magnetic induction near the centre of the solenoid is 0.08T. what is the current in the solenoid? 4
6.
 - a) What is electromagnetic induction? 1
 - b) State the Lenz's law. 2
 - c) The flux linking an air-cored coil of 500 turns changes from 30μWb to 60μWb in 2ms. Calculate the value of the emf induced in the coil. 4

8. Mathematical representation of Faradays law is also called

- | | |
|------------------|----------------------|
| a) Lenz's law | b) Biot-severt's law |
| c) Gauss's law | d) Ampere's law |
| e) Neumann's law | |

9. If the closed path does not link with the current loop, then the Ampere's law –

- | | |
|---|---|
| a) $\oint_l \vec{B} \cdot d\vec{l} = 0$ | b) $\oint_l \vec{B} \cdot d\vec{l} = \mu_0$ |
| c) $\oint_l \vec{B} \cdot d\vec{l} = i$ | d) $\oint_l \vec{B} \cdot d\vec{l} = n$ |
| e) $\oint_l \vec{B} \cdot d\vec{l} = BiL$ | |

10. What is the unit of emf?

- | | |
|------------|-----------|
| a) μF | b) TmA |
| c) Tm | d) $Volt$ |
| e) Am^2 | |

Invigilator Signature

Examiner Signature